# Skills Summary

## Writing Recursive Rules for Arithmetic Sequences

*Use this reference while completing your worksheet or when studying.*

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## Skill 1 — Finding the step and the starting value

**The recursive rule has two parts:** $a_1 =$ (starting value) and $a_n = a_{n-1} + d$ for $n \geq 2$. Neither part alone describes a sequence.

**Step from any two terms:** $d = \dfrac{a_k - a_j}{k - j}$ — divide by the number of *jumps*, which is one less than the number of terms involved.

**Walk back to term 1:** $a_1 = a_j - (j-1)d$.

**Example:** An arithmetic sequence has $a_3 = 14$ and $a_7 = 30$. Write its recursive rule.
**Step 1.** Two known terms fix the step, and the step walks the sequence back to term 1 — that is exactly the two pieces the rule needs.
**Step 2.** Terms 3 and 7 are four jumps apart, so the step is $\frac{30-14}{4}$.
**Step 3.** Term 1 is two jumps below term 3: $14 - 2(4)$.
$d = \mathbf{4}$, $a_1 = \mathbf{6}$, so $a_n = a_{n-1} + 4$ for $n \geq 2$.

**Try it:** $a_2 = 9$ and $a_6 = 29$. Write the recursive rule.

check: $d = \mathbf{5}$, $a_1 = \mathbf{4}$

## Skill 2 — Recursive rules from a situation

**Model:** a constant increase per stage is an arithmetic sequence. Decide what the index $n$ counts (rows, weeks, rungs) and what $a_n$ measures, with units.

**Then it is Skill 1 again:** the two reported stages are two known terms, so the same two formulas give $d$ and $a_1$. Check the rule against a reported stage before using it.

**Example:** A hall gains the same number of chairs each row. Row 3 holds 22 chairs and row 7 holds 34. Write the recursive rule.
**Step 1.** Equal chairs added per row makes the row counts an arithmetic sequence, so treat the two rows as two known terms.
**Step 2.** Rows 3 and 7 are four rows apart: $\frac{34-22}{4}$ chairs per row.
**Step 3.** Row 1 is two rows below row 3: $22 - 2(3)$.
$d = \mathbf{3}$ chairs per row, $a_1 = \mathbf{16}$ chairs, $a_n = a_{n-1} + 3$.

**Try it:** Row 2 holds 15 chairs and row 5 holds 27. Find the step and the first row.

check: $d = \mathbf{4}$, $a_1 = \mathbf{11}$

**(!) Watch out:** count *jumps*, not terms. From term 3 to term 7 is four jumps, and reaching term $n$ from term 1 takes $n-1$ jumps — the off-by-one here is the most common lost mark on this topic.

## Skill 3 — Extending and adding up

**Far-off terms:** applying the step $n-1$ times gives $a_n = a_1 + (n-1)d$, which is the same rule written so you can jump straight to a term instead of listing every one.

**Sum of the first $n$ terms:** $S_n = \dfrac{n}{2}(a_1 + a_n)$ — pair the smallest term with the largest.

**Example:** $a_1 = 7$ and $a_n = a_{n-1} + 6$. Find $a_5$ and the sum of the first 8 terms.

**Step 1.** Listing eight terms works but invites slips, so use the two shortcut forms the recursive rule generates.

**Step 2.** Term 5 is four jumps past the start: $7 + 4(6)$.

**Step 3.** Term 8 is $7 + 7(6) = 49$, so the sum pairs up as $\frac{8}{2}(7 + 49)$.

$a_5 = \mathbf{31}$ and $S_8 = \mathbf{224}$

**Try it:** $a_1 = 2$ and $a_n = a_{n-1} + 9$. Find $a_5$ and the sum of the first 6 terms.

check: $a_5 = \mathbf{38}$, $S_6 = \mathbf{147}$